

JEE MAIN REVISION MASTER: PRACTICAL ORGANIC CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 111 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Practical Organic Chemistry — Applied Cases, Reactions & Traps	84 PYQs	Applied Topics
Very High	Practical Organic Chemistry — Standard Calculations & Problem Archetypes	64 PYQs	Quantitative Analysis
High	Practical Organic Chemistry — Core Principles & Governing Laws	9 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Practical Organic Chemistry — Applied Cases, Reactions & Traps	$\times 10^{-1}g$, $\quad \times 10^{-1}$, $\quad \frac{0.80 \times 100}{1.60}$
Practical Organic Chemistry — Standard Calculations & Problem Archetypes	$\frac{10}{78}$, $\quad =\frac{4.5+4.5+4.4+4.4+4.4}{5}$, $\quad$ The value of y is _____. (Integer answer)
Practical Organic Chemistry — Core Principles & Governing Laws	$=\frac{4.5+4.5+4.4+4.4+4.4}{5}$, $\quad _{2}$, $\quad (N_{2})$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Practical Organic Chemistry — Applied Cases, Reactions & Traps (Very High Yield — 84 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Practical Organic Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Amongst the following, the total number of compounds soluble in aqueous NaOH at room temperature is :..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.481..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "Identify the correct statements : A. Hydrated salts can be used as primary standard. B. Primary standard should not undergo any reaction wit..." [Ans: N/A]

3.2 Practical Organic Chemistry — Standard Calculations & Problem Archetypes (Very High Yield — 64 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Practical Organic Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : A mixture of $C_{12}H_{22}O_{11}$ (sugar) and NaCl can be separated by dissolving sugar in al..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.481..." [Ans: B]
- [2026 Evening Shift] Q#2 (MCQ): "Complete combustion of X g of an organic compound gave 0.25 g of CO_2 and 0.12 g of H_2O . If the % of carbon is 25% and of ..." [Ans: A]

3.3 Practical Organic Chemistry — Core Principles & Governing Laws (High Yield — 9 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Practical Organic Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Practical Organic Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Practical Organic Chemistry PYQs.

Representative Source PYQs:

- [2026 Morning Shift] Q#3 (MCQ): "Given below are two statements : Statement (I) : 1,2,3-Trihydroxypropane can be separated from water by simple distillation. Statement (II) ..." [Ans: N/A]
- [2025 Evening Shift] Q#4 (Numerical): "In Dumas' method 292 mg of an organic compound released 50 mL of nitrogen gas (N_2) at 300 K temperature and 715 mm Hg pressure. The per..." [Ans: 18]
- [2024 Evening Shift] Q#2 (MCQ): "Chromatographic technique/s based on the principle of differential adsorption is / are A. Column chromatography B. Thin layer chromatography..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 28th January Evening Shift (Q#1)

A student has been given 0.314 g of an organic compound and asked to estimate Sulphur. During the experiment, the student has obtained 0.4813 g of barium sulphate. The percentage of sulphur present in the compound is _____.
(Given Molar mass in g mol⁻¹ S: 32, BaSO₄ : 233)

- (A) 63.15%
- (B) 21.05% **[Correct]**
- (C) 48.24%
- (D) 42.10%

Correct Answer: B

Solution Analysis: Mass of compound taken = 0.314g Mass of BaSO₄ obtained = 0.4813g In BaSO₄, sulphur present per mole = 32g Molar mass of BaSO₄ = 233g mol⁻¹ So, mass of sulphur in 0.4813g of BaSO₄ is:
 $m(S) = 0.4813 \times (32 / 233)$
 $m(S) = 0.4813 \times (32 / 233) = 0.06608\text{g}$ (approx.) Percentage of sulphur in the compound:
 $\%S = (0.06608 / 0.314) \times 100 \approx 21.05\%$ Correct option: B) 21.05%

Benchmark Problem 2: JEE Main 2025 (Online) 7th April Evening Shift (Q#4)

In Dumas' method 292 mg of an organic compound released 50 mL of nitrogen gas (N₂) at 300 K temperature and 715 mm Hg pressure. The percentage composition of 'N' in the organic compound is _____ % (Nearest integer)
(Aqueous tension at 300K = 15mmHg)

Correct Answer: 18

Solution Analysis: Given: Volume of N₂, V = 50 mL Pressure, P = 715 mm Hg Temperature, T = 300 K Aqueous Tension at 300 K = 15 mm Hg Calculation: Correct the Pressure for N₂: $P_{(N_2)} = 715\text{mmHg} - 15\text{mmHg} = 700\text{mmHg}$ Convert the pressure from mm Hg to atm: $P_{(N_2)} = (700 / 760)\text{atm}$ Calculate the Moles of N₂ using the Ideal Gas Law:
 $n_{(N_2)} = \frac{P_{(N_2)} \cdot V}{R \cdot T} = \frac{(700 / 760) \times (50 / 1000)}{0.0821 \times 300}$ Calculate the Moles of N: $n_{(N)} = 2 \times n_{(N_2)}$ Calculate the Mass of N: $\text{Mass of N} = 2 \times n_{(N)} \times 14$

Benchmark Problem 3: JEE Main 2024 (Online) 30th January Morning Shift (Q#3)

On a thin layer chromatographic plate, an organic compound moved by 3.5cm, while the solvent moved by 5cm. The retardation factor of the organic compound is _____ $\times 10^{-1}$.

Correct Answer: 7

Solution Analysis: Retardation factor = (Distance travelled by sample/organic compound / Distance travelled by solvent)
 $= (3.5 / 5) = 7 \times 10^{-1}$